

Robinson George Arysseril

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EDUCATION

Vimal Jyothi Engineering College

B.Tech in Artificial Intelligence and Data Science CGPA: 9.46 / 10

- Minor in Electronics and Communications

Chemperi, Kerala

Sep. 2024 – May 2028

TECHNICAL SKILLS

Data Science: Pandas, NumPy, Matplotlib, Scikit-Learn, SciPy, TensorFlow, LSTM, MinMax Scaling

Data Engineering: SQL, JSON, REST APIs, yfinance, structured database design

General Purpose: Python, Java, JavaScript, TypeScript, MySQL

Systems & Tools: Rust, C/C++, Zig, Git, Make

PROJECTS

Stock Market Price Predictor 🧠 | *Python, TensorFlow, NumPy, Scikit-Learn, Matplotlib* Feb. 2025 – Mar. 2025

- Built an end-to-end time-series forecasting pipeline: ingested historical OHLCV data for multiple companies via Yahoo Finance, applied MinMax scaling for normalization, and trained stacked LSTM layers to capture long-range temporal trends
- Produced 30-day forward price predictions per company, persisting results as structured JSON and validating model behavior through training/validation loss tracking
- Implemented incremental model training with checkpoint persistence, enabling continuous learning across sessions without restarting from scratch
- Developed Matplotlib visualizations comparing actual vs. predicted close prices and loss curves, providing clear model diagnostics and interpretability

News Bias Classifier (BABE Dataset) | *Python, PyTorch, Sentence-BERT, Scikit-Learn* 2025

- Built a binary NLP classifier on the BABE (Media Bias Annotations by Experts) dataset, distinguishing biased from neutral news headlines with **94% accuracy** on the held-out test set
- Generated contextual headline embeddings using DeBERTa-v3-large via Sentence Transformers and trained a custom 5-layer PyTorch network (1024→512→256→128) with dynamic dropout
- Designed a dual-objective training regime combining cross-entropy and contrastive loss with learnable log-space weighting parameters, optimized jointly with the model via Adam
- Implemented a neutrosophic logic classifier using cosine similarity to class centroids, computing truth, indeterminacy, and falsity membership scores for bias/neutral decisions
- Validated with 5-fold stratified cross-validation and a ReduceLROnPlateau scheduler with gradient clipping for stable convergence

One-Click Driver Installer & Hardware Database 🧠🧠 | *Rust, JSON, SQL* Sep. 2024 – Oct. 2024

- Designed and maintained a structured SQL and JSON database cataloging chipset metadata and driver mappings across 200+ Chromebook models, demonstrating end-to-end data modeling and schema design
- Dynamic runtime update mechanism fetches and applies the latest hardware metadata without application rebuilds, ensuring data freshness across a 5,000+ user install base

Chrultrabook Tools 🧠 | *Rust, Tauri, Angular, TypeScript* May 2023 – Present

- Cross-platform desktop application collecting and displaying real-time hardware sensor telemetry (fan speeds, temperatures, backlight levels) across Windows, macOS, and Linux; 45,000+ downloads
- Rust backend performs high-frequency sensor polling and fan curve calculations, exposing hardware data to the UI layer via IPC for live monitoring dashboards

Various Emulators | *Rust, C++, Zig* Oct. 2024 – Present

- **Game Boy (DMG) 🧠:** Cycle-accurate hardware simulation implementing CPU, PPU pipeline, MBC mappers, scanline renderer, OAM DMA, and sprite compositing — demonstrating strong computational modeling and state machine design
- **Intel 8080 🧠 / XO-CHIP 🧠 / CHIP-8 🧠🧠:** Multiple emulators across Rust, C++, and Zig covering full opcode sets, memory models, interrupt timing, and framebuffer rendering pipelines

EXPERIENCE

Open Source Contributor

May 2023 – Present

Chrultrabook Project

Remote

- Maintained Chrultrabook Tools, a cross-platform system utility serving users running alternate operating systems on Chromebook hardware
- Managed hardware compatibility database entries, ensuring accurate chipset and driver data across a growing catalog of device models
- Collaborated with project maintainers to improve documentation and user-facing tooling based on community feedback